



Factors and Multiples

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Common Multiples

Multiples that are common to two numbers are known as common multiples of those numbers. Let us understand with the help of an example.

Consider two numbers– 30 and 45. Multiples of 30 and 45 are –

30 = 30, 60, 90, 120, 150, 180, 210, 240, 270.....

45 = 45, 90, 135, 180, 225, 270.....

Common Factors

Factors that are common to two or more numbers are known as their common factors. So how do we find the common factors? Consider two numbers A and B. Write down all the factors of A and B separately and observe the numbers which are common to both A and B. Let us understand the concept of common factors with the help of an example.

Factors of 30 and 45 are –

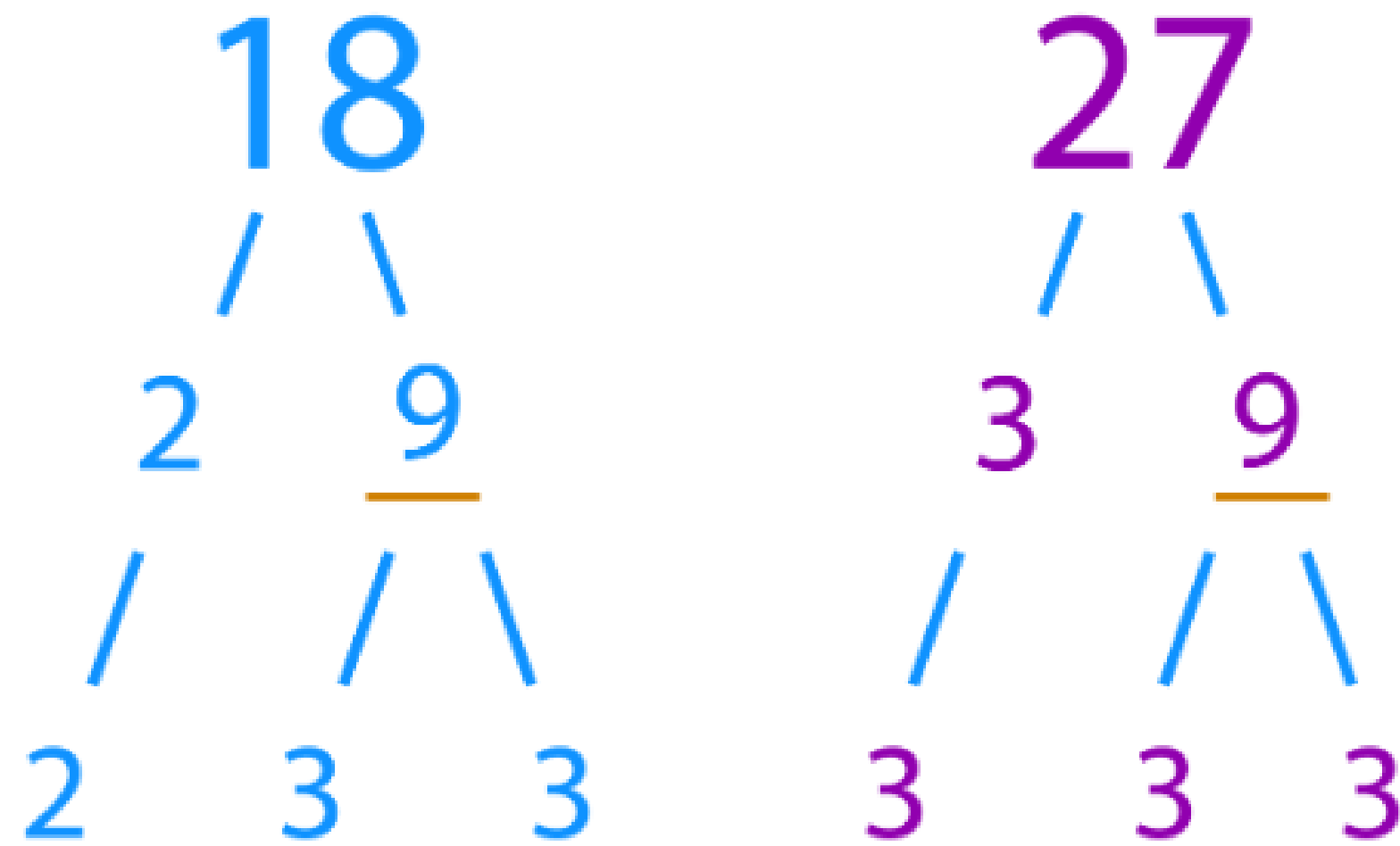
$$30 = \{1, 2, 3, 5, 10, 15, \text{ and } 30\}$$

$$45 = \{1, 3, 5, 9, 15, 45\}$$

What are the common factors that you can observe? 15, 5, 3 and 1 appear in both 30 and 45.

Common factors of 30 and 45 are 1, 3, 5 and 15.

What is the highest common factor between 18 and 27?



From this we can tell the HCF is 9.

Lowest Common Multiples (LCMs)

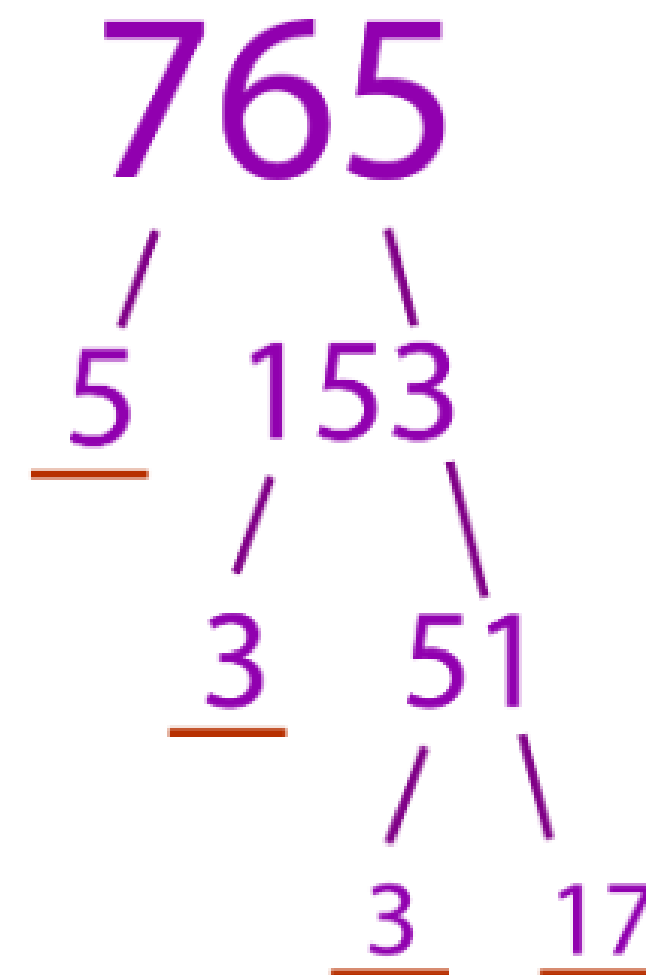
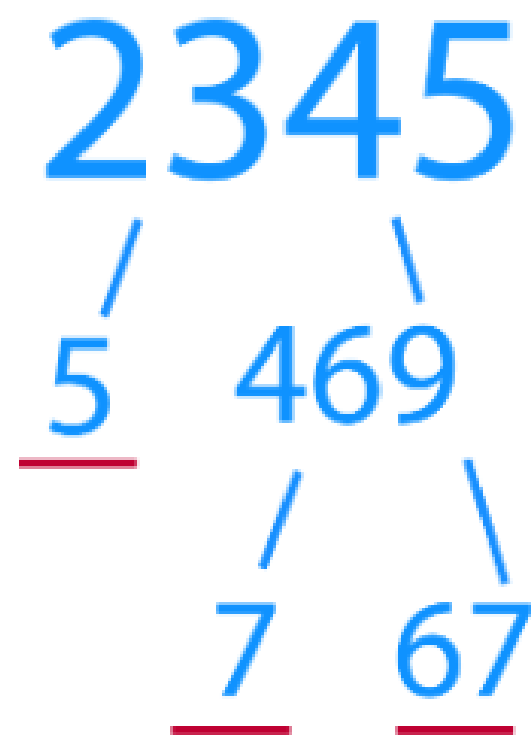
We can also use these prime factors to find the lowest common multiple between numbers.

Example

QUESTION 4

Evaluate: What is the lowest common multiple between 2 345 and 765?

Think:



The lowest common multiple is the smallest product of prime factors that contains all the prime factors of the two numbers. If there are common prime factors between the two numbers, you only have to include them once.

Do: $\text{LCM} = 5 \times 7 \times 67 \times 3 \times 3 \times 17 = 358\,785$